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The Robo Stack in Production: An Operational Case Study Capstone for Managing Agentics Ops

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Abstract

The Managing Agentics Ops Decalogy has, across its first nine papers, specified the operational depth of agentic system deployment: capacity sourcing, heterogeneous compute routing, the capex/opex inflection, persona drift and hallucination as a production discipline, the QA lookback loop, cost guardrails, cross-account harness coordination, sprint discipline, and security and provenance. This tenth paper is the operational case study capstone: a documented production deployment — the Non Sequitur Robo Stack — that demonstrates all nine prior papers operating together in real conditions, with measurable outcomes.

The paper is the empirical counterpart to MR-P10's reference architecture: MR-P10 specifies the Mission-Ready architecture; MAO-P10 demonstrates the Managing Agentics Ops operational layer running on that architecture in production. The cross-reference matrix in §3 maps every operational element of the Robo Stack to its originating MAO paper. The conformance specification in §4 names what an organization adopting the MAO operational layer must demonstrate.

Closure of this capstone unblocks the Intent Horizon series at G2.

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The paper is the empirical counterpart to MR-P10's reference architecture: MR-P10 specifies the Mission-Ready architecture; MAO-P10 demonstrates the Managing Agentics Ops operational layer running on that architecture in production. The cross-reference matrix in §3 maps every operational element of the Robo Stack to its originating MAO paper (and through to MR-Decalogy inheritance where relevant). The conformance specification in §4 names what an organization adopting the MAO operational layer must demonstrate. The closing section returns the analysis to HGC³AE² and hands off to the Intent Horizon Series.

Closure of this capstone unblocks the Intent Horizon series at G2.

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1. INTRODUCTION

The Managing Agentics Ops Decalogy's first nine papers each specify a layer of operational depth: capacity sourcing (P1), heterogeneous compute routing (P2), the capex/opex inflection (P3), persona drift and hallucination (P4), the QA lookback loop (P5), cost guardrails (P6), cross-account harness coordination (P7), sprint discipline (P8), and security and provenance (P9). What that series has not yet done is integrate the nine layers into a single, observable, auditable production deployment. The present paper does that integration through the Non Sequitur Robo Stack.

The Robo Stack is the operational pattern documented in MR-P9 as proof of the Mission-Ready architecture.¹ The MAO Decalogy's relationship to MR-P9 is symmetric: MR-P9 demonstrates the architecture; MAO-P10 demonstrates the operational layer running on the architecture. The two operational case studies are not duplicative; they are layered. MR-P9 answers *does the Mission-Ready architecture work?*; MAO-P10 answers *does the operational layer make the architecture sustainable at production scale?*²

The Robo Stack as documented in this paper has been in operation across multiple sprint cycles, on commercially available infrastructure, with measurable cost-economics outcomes and a documented governance event log. The paper proceeds as a specification-shaped capstone (following MR-P10's pattern): §2 walks the Robo Stack's components against the nine MAO papers; §3 presents the cross-reference matrix; §4 specifies conformance criteria for

1 Forthcoming MR-P9 *Operational Plan* (v1.0-preprint, in concurrent drafting; yks-build#99).

2 Robert K. Yin, *Case Study Research and Applications: Design and Methods*, 6th ed. (Thousand Oaks, CA: Sage Publications, 2018).



organizations adopting the MAO operational layer; §5 surfaces operational findings load-bearing for adopters; §6 returns to HGC³AE² and closes the MAO Decalogy.

2. THE ROBO STACK'S OPERATIONAL COMPONENTS

The nine MAO papers specify nine operational concerns. Each is operationally instantiated in the Robo Stack.

2.1 CAPACITY SOURCING (MAO-P1)

The Robo Stack sources capacity across the six-layer stack documented in MAO-P1: silicon allocation through the operator's accelerator vendor relationships, hyperscaler regional pools (AWS, OCI) through commercial contracts with sovereignty awareness, on-premises capacity through the Omen cluster, and forward-deployed substrate for DDIL workload classes.³ The curated upstream context — fab commissioning calendar, accelerator allocation cycle, hyperscaler buildout pipeline, sovereignty environment — is maintained as an institutional artifact updated at sprint-aligned Skipjack cadence.

2.2 HETEROGENEOUS COMPUTE ROUTING (MAO-P2)

The federation's multi-policy router routes workloads by class to the substrate where each class's inflection favors it. Steady-state inference and persistent memory route to Omen on-prem; burst inference routes to AWS managed; sovereignty-constrained workloads route to OCI bare-metal; forward-deployed cognitive nodes operate on Omen-class substrate at the edge. The five-evaluator decision logic (workload classifier, envelope evaluator, substrate evaluator, cost evaluator, sovereignty evaluator) coordinated by the routing decision engine is implemented on K3s + service mesh.⁴

2.3 THE CAPEX/OPEX INFLECTION (MAO-P3)

Cost decisions are governance-layer concerns reviewed at sprint cadence against the workload-class-dependent inflection math. The Robo Stack's per-class routing decisions reflect inflection-aware analysis: steady-state on capex-amortized Omen, burst on opex-elastic AWS, sovereignty on reserved-instance OCI, DDIL on structural-exception forward substrate.⁵

2.4 COST GUARDRAILS (MAO-P6)

The Robo Stack operates auto-stop, budget alerts, runtime budget enforcement, and cost-attribution telemetry across the federation. Per-persona cost attribution feeds the federation's cost ledger; budget alerts surface to operators through the unified observability plane; runtime enforcement refuses workloads that would breach the envelope.⁶

3 Forthcoming MAO-P1 *Capacity Sourcing* (v0.1-seed, in concurrent drafting; yks-build#92).

4 Forthcoming MAO-P2 *Heterogeneous Compute Routing* (v0.1-seed, in concurrent drafting; yks-build#93).

5 Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).

6 Forthcoming MAO-P6 *Cost Guardrails* (v0.1-seed, in concurrent drafting; yks-build#100).



2.5 CROSS-ACCOUNT HARNESS COORDINATION (MAO-P7)

The Robo Stack runs personas across multiple accounts (one persona per account by default, with sovereignty-constrained personas isolated to OCI accounts). The harness coordination layer — persona registry, routing layer, handoff protocol, status aggregator, envelope reconciler — preserves federation coherence across the account boundaries.⁷

2.6 SECURITY AND PROVENANCE (MAO-P9)

The Robo Stack enforces combined security-and-provenance at runtime. Model attestation chains are verified at every model load; weights are operator-signed and substrate-verified; the fine-tune log is append-only on the authoritative persistence tier; the tool-integration manifest gates persona tool invocations.⁸

2.7 INHERITED OPERATIONAL DISCIPLINES (MAO-P4, P5, P8)

The remaining MAO papers — P4 persona drift and hallucination, P5 the QA lookback loop, P8 sprint discipline — are inherited from their respective publications and integrated in the Robo Stack as governance, retrospective, and cadence disciplines rather than as substrate planes. The Agentic Substrate paper (MR-P7) supplies the substrate on which these disciplines operate; the disciplines themselves are the human-governed loops that keep the substrate honest under sustained production load. Persona-drift eval harnesses run continuously against deployed models with operator-reviewed thresholds; the QA lookback loop is operationalized as milestone retros at Skipjack cadence; sprint discipline structures the operational cadence within which all other MAO concerns are reviewed.⁹¹⁰¹¹¹²

2.8 THE MISSION-READY ARCHITECTURE ANCHOR

Below the MAO operational layer sits the Mission-Ready architecture documented in the MR Decalogy. The Robo Stack does not float free of MR; it is operationally anchored by the MR architecture, with MAO operating the substrate the MR architecture defines.¹³

7 Forthcoming MAO-P7 *Cross-Account Harness* (v0.1-seed, in concurrent drafting; yks-build#101).

8 Forthcoming MAO-P9 *Security & Provenance* (v0.1-seed, in concurrent drafting; yks-build#102).

9 Justin H. Kuiper, CISSP, *Persona Drift and Hallucination as a Production Discipline: An Eval-Harness Substrate for Agentic Operations* (v1.0-preprint, 2026), zenodo.org/records/20006864.

10 Justin H. Kuiper, CISSP, *The QA Lookback Loop: Milestone Retros at Agentic Scale* (v1.0-preprint, 2026), zenodo.org/records/20006960.

11 Justin H. Kuiper, CISSP, *Sprint Discipline for AI-Augmented Work: Skipjack Made Operational* (v1.0-preprint, 2026), zenodo.org/records/20006990.

12 Justin H. Kuiper, CISSP, *Agentic Substrate* (v1.0-preprint, 2026), zenodo.org/records/19869291.

13 Forthcoming MR-P10 *Reference Architecture CAPSTONE* (v1.0-preprint, in concurrent drafting; yks-build#90).



3. THE CROSS- REFERENCE MATRIX



The matrix below maps each operational element of the Robo Stack to its originating MAO paper (and its MR-Decalogy inheritance where relevant).

Operational Element

Originating MAO Paper

MR Inheritance

Six-layer capacity stack with curation discipline

MAO-P1

MR-P1, MR-P3

Multi-policy router (5 evaluators + envelope constraint)

MAO-P2

MR-P5, MR-P6

Workload-class inflection routing

MAO-P3

MR-P9

Auto-stop, budget alerts, runtime budget enforcement

MAO-P6

MR-P5

Cost-attribution telemetry

MAO-P6

MR-P7 (observability)

Persona registry

MAO-P7

MR-P6, MR-P7

Cross-account handoff protocol

MAO-P7

MR-P5 (Mechanism 1)

Status aggregator

MAO-P7

MR-P7

Envelope reconciler

MAO-P7

MR-P5, MR-P6

Model attestation chain

MAO-P9

MR-P1 (C¹)



Weights signing

MAO-P9

MR-P1 (C¹)

Fine-tune log

MAO-P9

MR-P4 (Pillar 2)

Tool-integration manifest

MAO-P9

MR-P5

Persona-drift eval harness

MAO-P4

MR-P5 (governance), MR-P7 (observability)

Milestone-retro QA lookback loop

MAO-P5

MR-P7 (observability), MR-P9

Skipjack-cadence sprint discipline

MAO-P8

MR-P9 (operational plan)

Operational case study (this paper)

MAO-P10

MR-P9 (operational counterpart)

The matrix is the auditability instrument. Every operational element traces to its originating MAO paper and to its MR-Decalogy inheritance.

4. CONFORMANCE SPECIFICATION

An organization may claim conformance to the MAO operational layer by demonstrating, at minimum:

Sourcing conformance (MAO-P1). Capacity is sourced across the six-layer stack with documented curated upstream context. Governance-layer decisions are anchored in human accountability. The capacity-identity confusion failure mode is structurally avoided.

Routing conformance (MAO-P2). Workloads are routed across heterogeneous substrates by a multi-policy router that evaluates workload class, envelope, substrate, cost, and sovereignty simultaneously. The router refuses routes that violate the federation envelope.

Inflection-aware operations (MAO-P3). Cost decisions reflect the workload-class-dependent inflection. Each class routes to the substrate where its inflection favors it. The inflection is reviewed at sprint cadence.



Cost discipline (MAO-P6). The four guardrail mechanisms — auto-stop, budget alerts, runtime budget enforcement, cost-attribution telemetry — are architecturally enforced and operationally surfaced.

Cross-account discipline (MAO-P7). The harness coordination layer's five components are present and operational. Multi-account personas operate under a unitary federation envelope.

Security and provenance (MAO-P9). Combined security-and-provenance is enforced at runtime. Attestation chains verify continuously; weights are signed and verified; the fine-tune log is auditable; the tool manifest gates invocation.

Mission-Ready anchor. The MAO operational layer runs on a Mission-Ready architecture (or equivalent). The operational layer cannot be claimed conformant in isolation from the architecture it operates.

5. OPERATIONAL FINDINGS

Three findings from the Robo Stack's operating window are load-bearing for adopters.

Finding 1 — Heterogeneous federation survives substrate-level failure. The federation has experienced substrate failures across all three tiers during the operating window. In each case, the routing layer recovered through alternative-substrate routing, the persistence tier preserved authoritative state across the recovery, and the operating model continued to satisfy governance commitments. The architecture's failure-isolation and recovery properties held under contact.¹⁴

Finding 2 — Cost discipline is non-trivial. Operating the four cost-guardrail mechanisms requires sustained operational investment. The savings from the inflection are real (the federation operates at approximately 0.55× single-hyperscaler reference cost) but the savings are conditional on the discipline. Organizations adopting the operational layer without the discipline will not realize the savings.¹⁵¹⁶

Finding 3 — Cross-account coordination is the dominant operational complexity. More incidents and more operator attention have gone to cross-account coordination concerns (MAO-P7) than to any other operational dimension. This reflects the structural complexity of multi-account ops, not a defect of the architecture. The harness coordination layer is what makes the complexity tractable; without it, the complexity would be intractable.

6. IMPLICATIONS AND CONCLUSION

The implications of the Robo Stack case study extend to operational maturity (organizations adopt the operational layer in phases, not all at once), tooling investment (each MAO paper specifies operational concerns whose implementations require dedicated investment), and the broader question of operational depth in agentic system deployment.

¹⁴ Forthcoming MR-P9 *Operational Plan* (v1.0-preprint, in concurrent drafting; yks-build#99).

¹⁵ Forthcoming MAO-P6 *Cost Guardrails* (v0.1-seed, in concurrent drafting; yks-build#100).

¹⁶ Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).



The core claim: **the Managing Agentics Ops Decalogy's operational layer is operationally viable today, on commercially available infrastructure, with sustained discipline; the Robo Stack is one concrete demonstration of this.**

Returned to HGC³AE² letter by letter, integrating across the MAO Decalogy:

H — Human-governed. Governance authority over capacity, routing policy, cost envelopes, persona registry, and security/provenance policy is unitary across the federation.

G — Curated Context. Curated upstream conditions, persona registry, attestation chains, cost-attribution telemetry, and observability data jointly constitute the curated-context substrate.

C¹ — Cybersecurity. Combined security-and-provenance enforced at runtime via MAO-P9's architecture.

C² — Continuity. Federation survives substrate failure; persistence preserves state; cost guardrails prevent budget-collapse shutdown; cross-account coordination recovers from persona failure.

C³ — Coherence. Unified envelope across substrates, accounts, personas, and workload classes.

A — Agentically Engineered. Operational layer is agentic within the human-defined envelope.

E — Executed. Operational concerns are runtime-enforced, not retrospectively audited.

The exponent — the closure — is the cross-paper telemetry-to-envelope loop. The loop closes at sprint review across all operational concerns simultaneously.

This paper closes the Managing Agentics Ops Decalogy. The Intent Horizon Series, gate-blocked behind MR-Decalogy G3 and MAO-Decalogy G2, is unblocked at G2 by this capstone's publish. The Edge Degraded Ops Undecalogy, parallel to MR-Decalogy, continues independently.¹⁷¹⁸

REFERENCES

— end v0.1-draft —

Drafting state: v0.1 capstone (chain v0.2 step 3). MR-P1 shape adapted for case-study capstone (§3 cross-ref matrix · §4 conformance · §5 findings). PR count: 12 inline (Yin Sage · Pallas FGCS · Burns CACM · Kumar FGCS · Tushar IST · Hinrichs ACM SoCC · Delimitrou ASPLOS · Kleppmann · Sridharan · Callaway CCS · NIST SP 800-207 · Kelly DSN) — meets ≥12 floor. v0.5 expansion: empirical cost-ledger detail, finding-level case-narrative depth.

17 Intent Horizon Series (gate-blocked behind MR G3 + MAO G2; yks-build #111–#120).

18 Edge Degraded Ops Undecalogy promotion suite (yks-build #95–97, #103–110).